

Microeconomic Analysis: Problem Set 3

Static Optimisation

Submit questions 3,4,6

January 31, 2026

1. Find all the critical points (solutions to the first-order conditions) of the function

$$f(x, y) = x^4 + x^2 - 6xy + 3y^2,$$

and classify each one (if possible) as a local maximum, local minimum, or saddle point.

2. Show that the function

$$f(x, y) = x^2 + 2y^2 + xy + 3x + 19y - 4$$

is strictly convex, by examining the determinant and trace of the Hessian and then using the eigen-value test, or otherwise.

Find the pair (x, y) that minimises f .

3. Use the Lagrangian method to find maximum and minimum values of

$$x^2 + y^2 \quad \text{subject to} \quad x^2 + xy + y^2 = 3,$$

checking the constraint qualification and the second-order conditions.

4. A consumer obtains utility $u = x^2y$ from consuming quantities x and y of goods X and Y . The price of good X is 2, the price of good Y is 3, and his total income is 9. He chooses x and y to maximise his utility, subject to his budget constraint:

$$\max_{x,y} x^2y \quad \text{subject to} \quad 2x + 3y \leq 9, \text{ and } x \geq 0, y \geq 0$$

- (i) Explain why his choice must satisfy $2x + 3y = 9$, and $x > 0, y > 0$.
- (ii) Write down the Lagrangian for the problem, and solve the first order conditions.
- (iii) Show by checking the second-order conditions that the solution you have found is a strict local maximum.
- (iv) Explain why this must also be the global maximum.

5. A firm produces output $y > 0$ from inputs K and L . Its production possibilities are constrained by $K \geq 0, L \geq 0$ and $Y(K, L) \geq y$ where

$$Y(K, L) = (K + 1)^{1-\alpha}(L + 1)^\alpha - 1 \quad \text{with } 0 < \alpha < 1.$$

If the firm takes r and w , the strictly positive prices of K and L , as given, then what is the least-cost way of producing y , and what is this cost?

(Give bounds(s) on the ratio r/w in terms of α and y under which the solutions you have found are valid.)

6. A household has preferences represented by the utility function $u(x_1, x_2)$ where

$$u(x_1, x_2) = x_1^\beta + x_2^\beta \quad \text{with } 0 < \beta < 1;$$

it is constrained by $x_1 \geq 0, x_2 \geq 0$ and also by

$$p_s x_1 + x_2 \leq p_s m_1 + m_2, \quad p_b x_1 + x_2 \leq p_b m_1 + m_2$$

where m_1, m_2 are fixed endowments, both strictly positive. The price of good 2 is normalised to 1, and we also have $0 < p_s < p_b$.

Draw the budget set. What is the interpretation of p_s & p_b ?

What choices should the household make if it wants to maximise utility?

(Give bounds(s) on p_s and/or p_b in terms of β, m_1 and m_2 under which the solutions you have found are valid.)